Asjad Akhtar

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WORK EXPERIENCE

Kalam Labs — **Lucknow, UP**
Software Engineer — *December 2023 - January 2025*

- Architected and implemented the full-stack company website using React.js, Node.js (Express), and MongoDB, delivering a responsive UI and scalable RESTful APIs to support 5,000+ monthly users.
- Designed and enforced MongoDB schema models and data-access layers, optimizing read/write performance by 30% and ensuring ACID-like consistency across user, orders, and telemetry collections.
- Integrated payment gateways (Razorpay, Stripe) via secure, PCI-compliant REST endpoints; automated webhook handlers to process transactions, refunds, and subscription lifecycles with idempotent operations.
- Containerized back-end services with Docker, set up CI/CD pipelines (GitHub Actions) to automate build, test, and deployment workflows, cutting release cycle time by 40%.

Salesqueen Software Solutions — **Chennai, TN**
Front-End Intern — *June 2023 - July 2023*

- Implemented and optimized key CRM user interfaces using React, Tailwind CSS, and vanilla JavaScript, refactoring legacy jQuery components to React, reducing front-end codebase by 25%, and improving page-load times by 15%.
- Integrated reusable React components (modals, data tables, form validations) and managed state with React Hooks, accelerating feature delivery cycles by 30%.
- Wrote comprehensive unit tests with Jest and React Testing Library, achieving 80% front-end test coverage and reducing post-release UI bugs by 40%.

PROJECTS AND PUBLICATIONS

NES Emulator from Scratch [GitHub]
Personal Project — *May 2025*

- Architected and implemented a cycle-accurate NES emulator in C++ using object-oriented design principles, modeling CPU (6502), PPU, and APU subsystems, achieving high instruction-level accuracy..
- Developed memory-mapped I/O, sprite rendering, and audio waveform generation engines, enabling flawless playback of 20+ classic NES titles (e.g., Super Mario Bros., The Legend of Zelda) at a steady 60 FPS.
- Designed modular, reusable C++ classes abstracting CPU registers, PPU memory tables, and APU audio channels; reduced code complexity by 30% and facilitated future feature extensions (e.g., MMC mapper support).

Anime Tracker App Using Flutter [GitHub]
Personal Project — *February 2025*

- Developed a cross-platform Flutter application integrating with the MyAnimeList API to enable users to browse seasonal anime, mark titles as watched, and curate a personal watchlist, handling CRUD operations across three distinct lists.
- Engineered state management with Provider and implemented Hive-based local storage to cache up to 200 anime entries for offline access, reducing network calls by ~40% and improving initial screen load times by ~30%.
- Integrated Dio for performant, authenticated RESTful requests (using MyAnimeList client ID), complete with retry logic and error-handling flows that maintained 99% API uptime for end users.

5G Embedded Decentralized Systems for Secure and Efficient Data Sharing
Book Chapter (First Author) || [Link] — *July 2024*

- Conducted an in-depth analysis of 5G network architectures to identify ultra-low latency and high-throughput data exchange mechanisms in decentralized edge systems.
- Designed and evaluated encryption schemes leveraging 5G's native security features (e.g., 5G-AKA, subscriber authentication) to ensure end-to-end privacy and integrity across peer-to-peer data channels.

Hybrid Localization and Navigation System for Dynamic Target Employing RSSI and A* Algorithm
Journal Paper (First Author) || [Link] — *February 2024*

- Developed a hybrid indoor localization/navigation framework in response to an ISRO SIH 2022 problem statement; achieved finalist status (among 50+ teams) in the ISRO SIH 2022 hackathon.
- Validated through simulation and prototype testing, demonstrating robust real-time position estimates and optimal route computation in complex indoor layouts.

EDUCATION

Amity University — **Lucknow, UP**
B.Tech Computer Science and Engineering — *July 2024*

TECHNICAL SKILLS

Languages: C++, C, C#, Python, Java, JavaScript, Dart

Web & Backend: React, Node.js, Express, REST APIs

Databases: SQL (MySQL/PostgreSQL), NoSQL (MongoDB)

DevOps & Tools: Docker, Git, CI/CD, Unix/Linux